

REVERSE ENGINEERING

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ENSET-M

Bootcamp Plan

01

Intro

02

**Linux
binaries**

03

**Windows
binaries**

04

**Android
binaries**

- **How programs work**
- **Compilation process (source → compiler → binary)**
- **File Formats**
- **What is reverse engineering**
- **Static vs dynamic**
- **Tools**
- **What can block us ?**
- **Labs**

What Is a Program?

A program is a set of instructions that tells a computer exactly what to do, step by step, to perform a specific task.

What Is a Program?

What Is a Program?



What Is a Program?



What Is a Program?



Compilation process

Compilation process

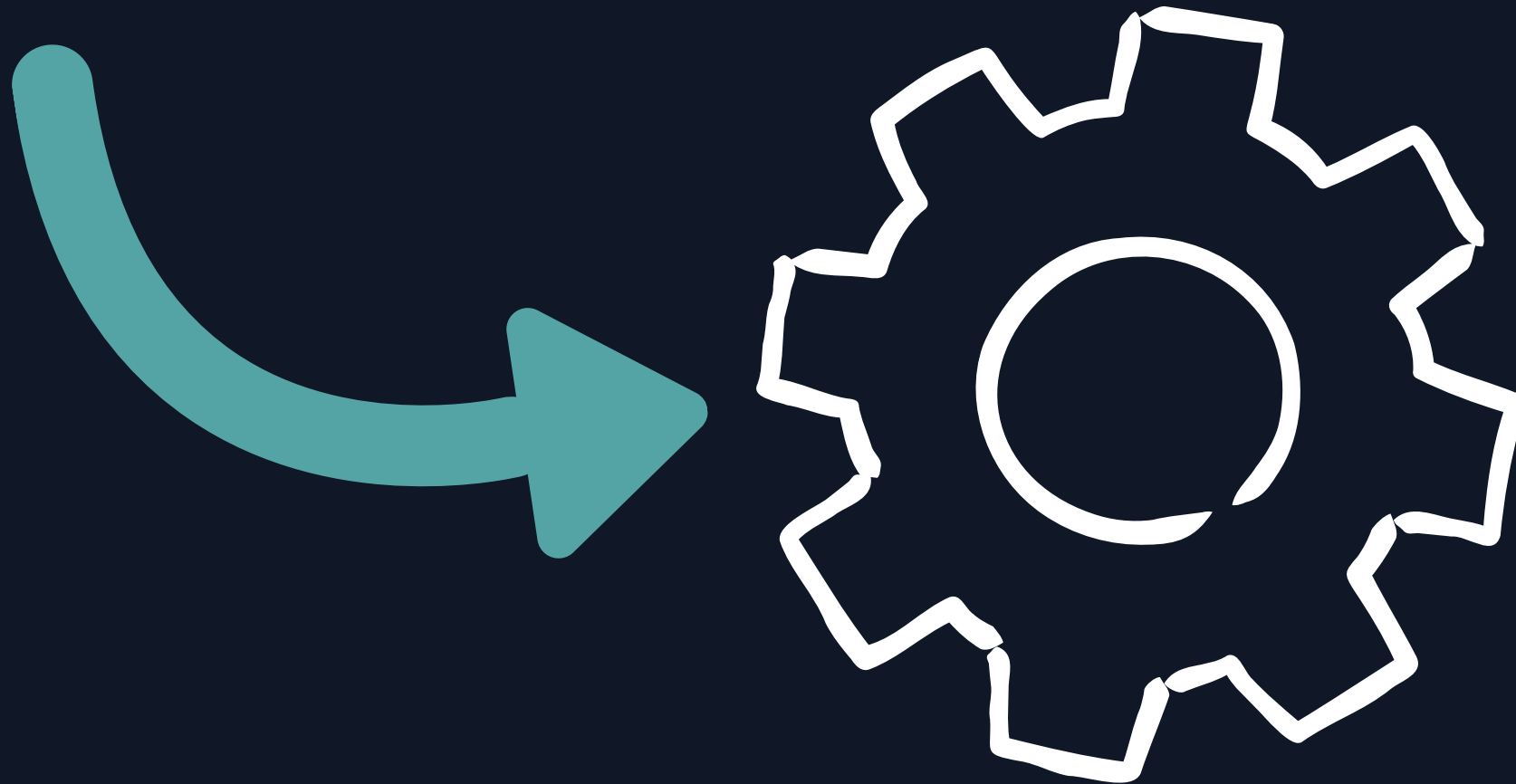
```
#include <stdio.h>
```

```
int main() {  
    printf("Hello, World!\n");  
    return 0;  
}
```

Compilation process

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int main() {  
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Compilation process

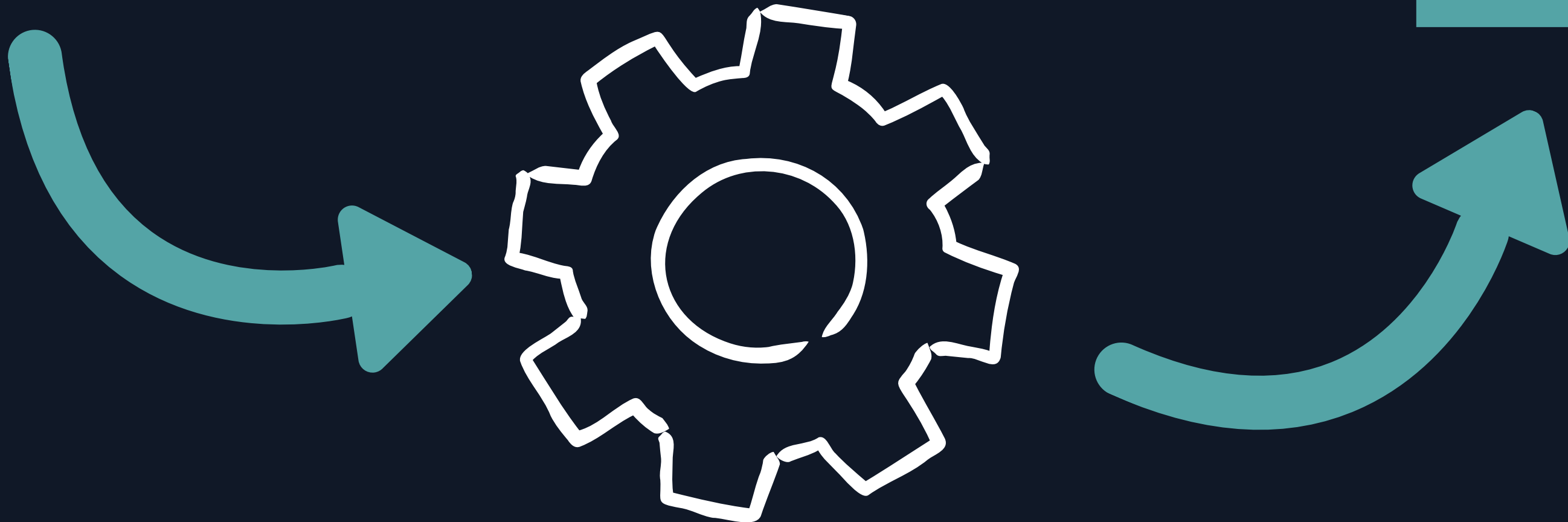
```
#include <stdio.h>
```

```
int main() {  
    printf("Hello, World!\n");  
    return 0;  
}
```

0101

0101

0101



Compilation process





ach taygol had
khona ?

```
printf("I love Pizza");
```





khoya ana kanbghi
tacos

00100000	01110000	01110010
01101001	01101110	01110100
01100110	00101000	00100010
01001001	00100000	01101100
01101111	01110110	01100101
00100000	01010000	01101001
01111010	01111010	01100001
00100010	00101001	00111011



File Formats



Reverse engineering

Understanding how a program works without having its source code.

Why RE?

- Understand program behavior
- Find vulnerabilities
- Recover lost source code
- Malware analysis

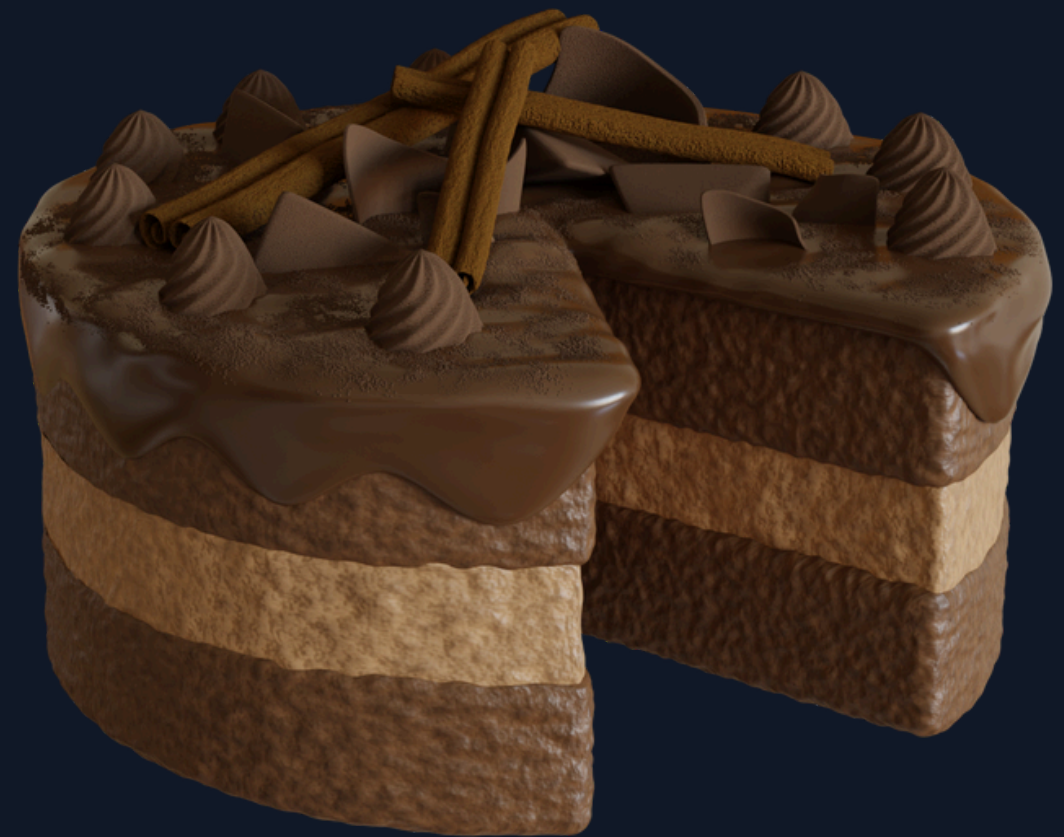
Static vs Dynamic

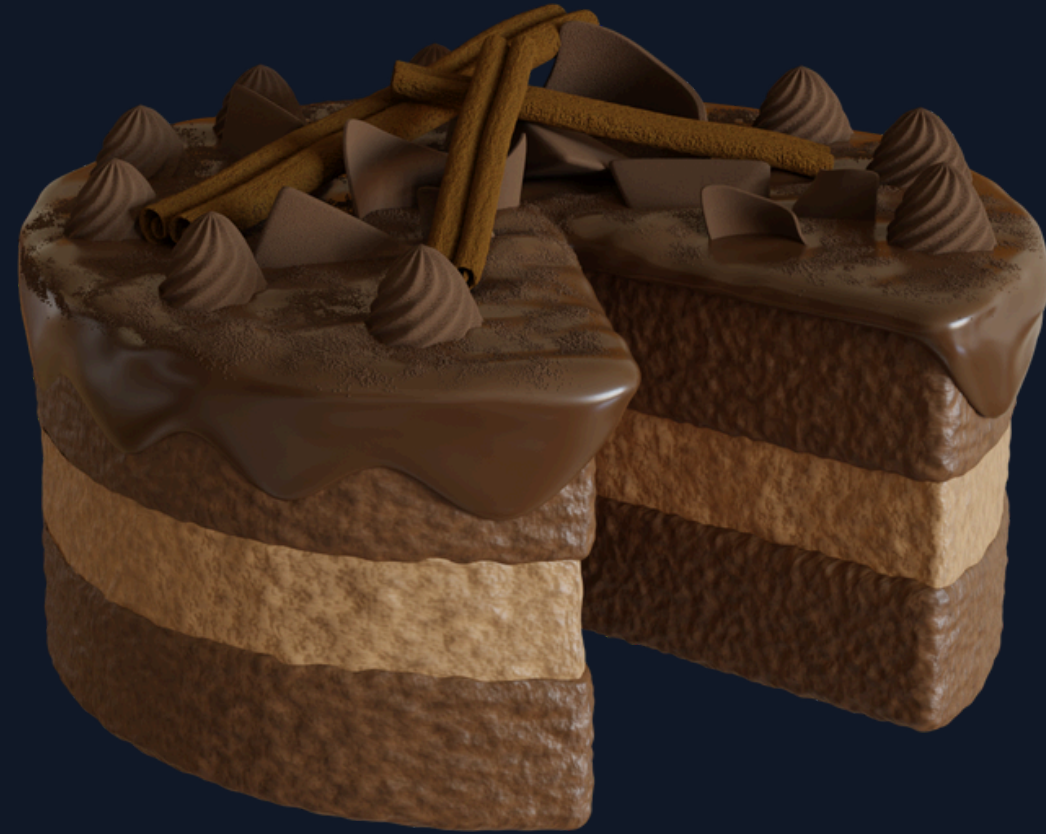


Looking inside the program
without running it.

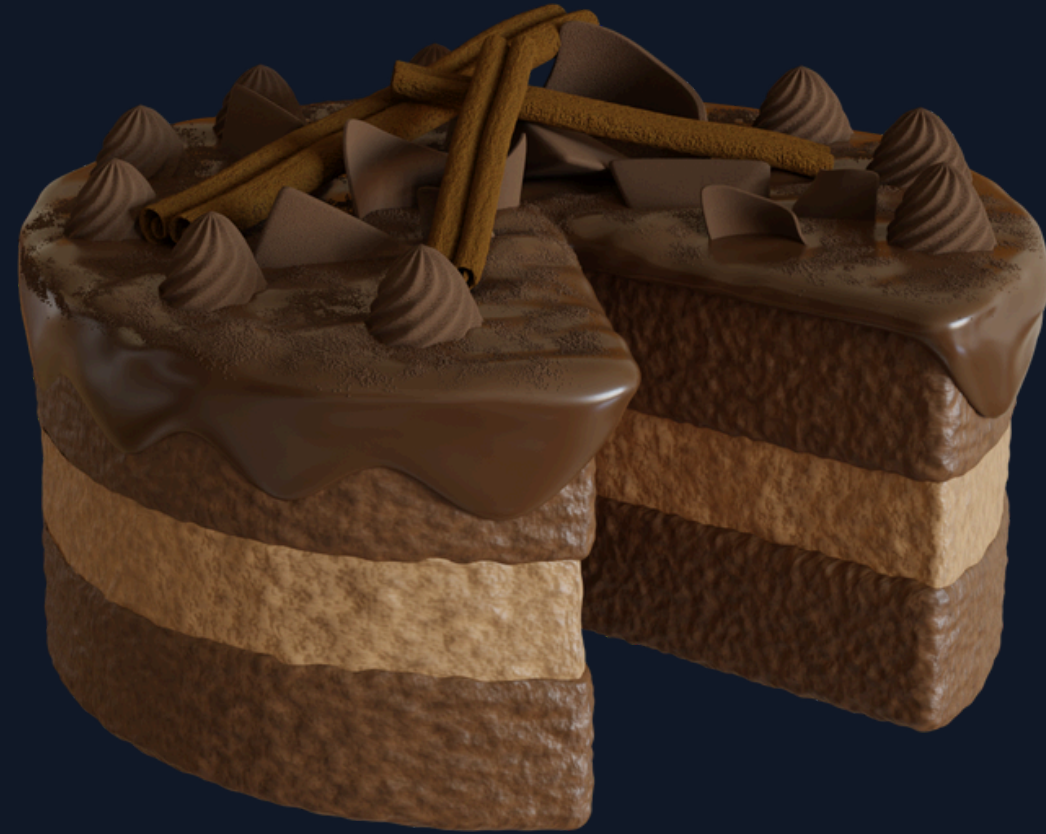


Observing what the program does
while it runs.





Looking



Looking



Tasting

Tools



objdump

strings



FRIDA

What can block us?

- **Stripped symbols**
- **Obfuscation / packing**
- **Encryption / checksums**
- **ASLR / PIE**
- **NX / DEP / Memory protections**
- **Anti-debugging**

Quick detection & what to try first



strings

checksec



Quick detection & what to try first



strings

checksec



Quick detection & what to try first



strings

checksec



THANK YOU